

Ontological Realist Reductionism

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DOI:10.5281/zenodo.22991102

Introduction

1. Philosophy understands the world, and the angles and ways of understanding differ; physics describes the world, but the ways of description are also not unique. The same set of phenomena can be organized and understood through different frameworks. Some frameworks excel at prediction, others at explanation. The instrumentalist position is: as long as predictions are correct, it does not matter whether there is an intelligible physical picture inside the framework. The position of ontological realism is different: it requires that a theory not only give numbers, but also give a clear picture of "what the world is like" and "how it evolves." Although human observational technology is limited, and the precision and range of observations and measurements of the universe are limited, we cannot give up attempts to understand the ontology of the universe, because only by continuously attempting to understand can we continuously improve our cognition of the universe. If we give up understanding the ontology of the universe and are merely satisfied with instrumental calculation, we may fall into the trap of partial and one-sided cognition, because without conjectures about ontological reality, it is actually difficult to falsify, truly falsify, and actual cognition is also difficult to progress.

2. This article takes a specific model that the author began developing in February 2025 as an example to explain how, based on phenomena and in accordance with the philosophical conviction that our universe is a real subject, to construct a worldview consistent with phenomena. This set of ontological commitments is certainly not an exact representation of the origin of the universe, but the author believes that only by giving clear ontological commitments can cognitive improvement be obtained in the process of continuously having ontological cognition corrected by phenomena.

3. The core claim of this model is: space is real, and from actual observation, any real object cannot be infinitely divisible; there should exist a bottom-level basic composition. Only through the method of reductionism, tracing back to the bottom level, even if this bottom level cannot be

observed, and forming a normalized architecture starting from the bottom level through reasoning and conjecture, can concepts at different levels such as time, entropy, mass, and gravity in the universe be represented in a simple and understandable way.

4. Another philosophical claim of the author is: our observation and perception of the universe is 3D, so if the temporal evolution of a 3D universe can be used to characterize changes in the universe, then there is no need to use higher-dimensional models to understand our universe, because the higher the dimension, the more detached from intuition, and the more detached from the range of falsifiability.

5. Let us try to see whether the temporal changes of 3D space can cover the main physical phenomena.

6. Each section of this article will explain: how that section, taking ontological realist reductionism as its principle and aiming to give a 3D real physical process picture, and what advantages in physical picture this way of working forms.

I. Starting from phenomena, form a bottom-level ontological real object in the cognitive sense.

7. Ontological commitment is the starting point of cognition. This starting point may be wrong, but without this starting point, even being wrong cannot be discussed. Humanity's understanding of the universe has always proceeded from phenomena to ontological real cognition, and then new phenomena are discovered to revise previous ontological real cognition. Principles and mechanisms of operation are established on the basis of ontological real objects, and relations are also established on the basis of objects. If ontological objects are not defined, relations and calculations are actually floating and unsettled. I personally believe that even the masters of contemporary computationalism actually also have their own ontological real understanding of the universe, because I think that without an internal ontological real definition, it is actually impossible to carry out process deduction, let alone explore the principles of its evolution. Contemporary science, including physics, actually cannot form a self-contained system detached from philosophical ontological cognition. Contemporary science's contempt for philosophy on certain occasions is actually not objective enough.

8. The first step of ontological realist reductionism is to determine which phenomena need to be explained. This is not an arbitrary choice, but starts from phenomena that are observable, repeatable, and do not depend on a specific theoretical framework.

9. In the process of cosmic evolution, time flows and is irreversible; this is the most direct experience. Heat flows from hot places to cold places; this is the daily manifestation of the second law of thermodynamics. Objects have mass, and masses attract each other; these are basic facts of mechanics and gravity. Particles have spin, charge, and magnetic moment; these are stable results of quantum experiments. These phenomena are the way the world presents itself to us.

10. The work of ontological realist reductionism at this step is: to regard these phenomena as "items to be explained" that need to be reduced, rather than treating them as independent basic facts. Instrumentalism can accept that "time is time" and "mass is mass," as long as the equations can be calculated correctly. But if one wants to form an internally logically consistent and comprehensive understanding, a bottom-level evolutionary mechanism is needed. It asks: is there a common ontology behind these phenomena? If so, what is this ontology?

11. *Space*, all physical processes are changes in space. According to existing achievements of general relativity and quantum field theory, space-vacuum is not empty; gravity is the deformation of vacuum, and microscopic particles are excitations of vacuum. Everything is a change in the state of space itself.

Space can be understood as an ontological real object, and all physical processes can be reduced to changes in space itself. This is our starting point.

12. **The advantage in physical picture formed at this step:** when phenomena are uniformly regarded as different manifestations of the same ontology, analogies and connections between them become natural. Time, entropy, mass, and gravity are no longer mutually unrelated concepts.

II. Reduce to the bottom level: normalized simple representation

13. The goal of reductionism is not simply to simplify complex phenomena, but to find the basic ontology and basic rules, to cover phenomena and form a unified understanding.

14. If all physical processes are the evolution of space itself, then what bottom-level components is space composed of, and are the adjacency relations between components changeable? How can it represent both the deformable space of general relativity and the quantization of quantum field theory?

15. Let us first discuss whether the adjacency relations between the basic components of space (quantum sites) are changeable. If the adjacency relations between spatial quantum sites can change, then the speed of light in vacuum would certainly be unstable, so the adjacency relations

between spatial quantum sites are more likely to be unchangeable.

16. Let us then discuss the macroscopic properties of space. If space is deformable, from an intuitive understanding, it is like an elastic body. Adding the property just discussed that adjacency relations are unchangeable, it is even more like an elastic solid. Note: understanding space as a deformable 3D elastic solid has been continuously studied by scholars since the 19th century. Since Clifford proposed in 1876 that matter and matter's motion themselves are only the propagation of spatial deformation, this program has always attracted many scholars, and this program is consistent with the phenomenon that the speed of light cannot be superimposed in the Michelson-Morley experiment.

17. At this point, the bottom-level ontology has only two elements: a fixed topological network, and energy values at the sites.

18. The dynamic rule of evolution: Clausius gave the original form of the second law of thermodynamics and entropy increase as evolution driven by thermodynamic gradients. This is thermodynamic intuition, although contemporary thermodynamics and other branches of physics have blurred this bottom-level drive of energy gradient driving, almost all phenomena including quantum processes support that the evolution of isolated systems follows gradient driving. Contemporary scholar Professor Annala explicitly proposed and insisted on this bottom-level driving rule in an article published in 2009. At this point, we can assume that the bottom-level rule of evolution on space is the conduction of energy from high to low between basic spatial sites, until we discover the existence of inverse-gradient phenomena in isolated systems. Of course, some scholars of quantum theory may hold different views on this; letters or public criticism and debate are welcome.

19. At this point, we have understood the ontological real object space and the bottom-level driving rule of its evolution.

III. The definition of time

20. In standard physics, time is usually treated as a basic parameter, denoted by the symbol t , but what it itself is is often not further explained. Ontological realism does not accept this attitude of "using but not explaining." It requires that time also be reduced to the bottom-level ontology.

21. The definition given by this model is: **time is the counting of state transformations of the spatial network.**

22. Then the concept of time becomes clear here. For example, time can be directly understood as the counting of transformations on this 3D deformable network whose topological adjacency relations are unchanged. The direction of time is the direction of evolution driven by gradients on this network. Under gradient driving, the energy distribution on space is irreversibly becoming more and more uniform. Later we will mention how to characterize this process of uniformization.

23. Specifically, each time energy transfers from a high-energy node to an adjacent low-energy node, it is one spatial transformation. Time is the counting of such transformations. It can be assumed that Planck time is the minimum interval between two adjacent transformations.

24. The work of ontological realist reductionism at this step is: to reduce time from an independent entity to a counting process of bottom-level network dynamics. Time is not a geometric dimension, but the counting of state transformations of this 3D geometric body, the universe.

IV. A non-subjective, non-statistical analytical definition of entropy

25. In standard thermodynamics, entropy usually has two major classifications of definitions. One is the thermodynamic quantity proposed by Clausius under temperature-gradient driving; the second is the logarithm of the number of microstates under the contemporary statistical paradigm, $S = k \ln W$. Although the statistical definition is successful, it depends on the "number of microstates," and the number of microstates often requires subjective choice of the degree of coarse-graining.

26. Ontological realist reductionism finds it difficult to accept entropy as a measure of subjective uncertainty. It requires that entropy should be an objective function of the evolution of the bottom-level ontology.

27. The author here proposes multiplicative entropy: as discussed in the previous sections, we understand the universe as a 3D physical network composed of a finite number of spatial quantum sites. Suppose this isolated system, the universe, has N sites (this number may be very large, but it is a natural number), and the i -th node carries energy m_i . Define

Total energy conservation: $\sum_i m_i = \text{constant}$

Entropy: $S = \prod_i m_i, i \in N$

28. When energy transfers from node i to node j , and satisfies the energy gradient rule $m_i > m_j + 1$, the ratio of entropy after transfer to entropy before transfer is

$$S_2 / S_1 = (m_i - 1)(m_j + 1) / (m_i m_j) = 1 + (m_i - m_j - 1) / (m_i m_j) > 1$$

29. Therefore every energy transfer (spatial transformation) increases multiplicative entropy. When all node energies are equal or differ by only one Planck constant, entropy reaches its maximum.

30. The work of ontological realist reductionism at this step is: to reduce entropy from a statistically vague quantity to a deterministic product. Entropy no longer needs the concept of "number of microstates," which requires subjective definition, but is directly calculated from the energy values of bottom-level nodes. This is the power of ontological realism and reductionism. As long as we reduce to the bottom level, we can describe physics in a simple way. Although this bottom-level object is unobservable, reasoning about the bottom-level ontology and evolutionary laws under the constraints of phenomena can improve our understanding of natural laws.

31. The advantage in physical picture formed at this step: entropy increase is no longer the vague statement that "the system tends toward disorder," but an energy redistribution process that can be tracked step by step. Each energy transfer corresponds to a definite change in entropy value. Entropy is no longer a measure of subjective uncertainty, but an objective function of the degree of uniformity of energy distribution. After taking the logarithm, multiplicative entropy is consistent in form with Boltzmann entropy, but the former is exact and step-by-step computable, while the latter is statistical and coarse-grained.

32. At this point, we can also understand spatial transformation, time, and entropy in a unified way: one spatial transformation corresponds to one moment, and one spatial transformation corresponds to one computable entropy value.

V. The elastic-geometric duality of mass-gravity

33. Mass and gravity are the two most mysterious concepts in standard physics. Mass is "the inherent quantity of matter," and gravity is "the geometric effect of spacetime curvature." Ontological realism requires that they too be reduced to the bottom-level ontology.

34. Based on the previous sections' discussion of the ontological reality of space analogized as an elastic solid, the reduction given by this model is: **mass is space being compressed, gravity is space being stretched.**

35. We continue along the program proposed by Clifford, W.K. in his 1876 article that matter itself is also spatial deformation. Combining achievements of contemporary quantum

field theory and condensed matter physics, if matter and matter's motion themselves are traveling wave packets on space, matter and gravity have the same bottom-level composition, and gravity is an attractive potential. From the elastic analogy, it is like the restoring potential of a stretched state, and mass can induce a gravitational field. Then mass itself is also spatial geometric deformation, and the essence of mass is very likely a spherically symmetric compression deformation of space. Imagine an elastic solid: after some local compression, the remaining part must be stretched. This picture is intuitive and logically self-consistent.

36. Here we see the advantage of ontological realist reductionism. If one adheres to the philosophical view of ontological realist reductionism, starting from simple phenomena and logical reasoning, one can deduce that the essence of mass is a compressed state of space. If one does not hold the philosophical standpoint of ontological realism, the understanding of mass will remain trapped in mathematical symbols and cannot form an intuitive understanding at the cognitive level.

VI. The advantages of ontological realist reductionism

37. From the above introduction, several advantages of the philosophical standpoint of ontological realist reductionism can be seen.

38. **First, the ontology is clear.** The model clearly answers "what the world is composed of": a network of spatial quantum sites with invariant topology. All other concepts are reduced to this ontology or to changes in this ontology. This clarity is not brought by mathematical formalization, but by ontological commitment.

39. **Second, the mechanism can be pictured.** For some mysterious concepts such as the essence of time and mass, there are understandable 3D physical pictures and mechanistic understandings. It no longer relies on "ineffable quantum probability" or "abstract spacetime curvature." Every concept can be imagined as something happening in space.

40. **Third, mathematical representation is normalized.** Time and entropy both become integer counts, and physics at the bottom level becomes arithmetic.

41. **Of course, the main effect is to escape mysticism.** The model does not introduce extra dimensions, does not introduce nonlocal interactions, and does not rely on statistical approximation. All phenomena emerge from the same bottom-level network and the same dynamical rule. This power to escape mysticism comes from the insistence of ontological realism

on "intelligibility."

42. Of course, although this set of understandings is conceptually consistent with most phenomena, it is still at the conceptual framework stage, with insufficient mathematical formalization and limited experimental verification. As a case of ontological realist reductionism, it demonstrates the explanatory power of this philosophical standpoint: when we seriously treat the two premises "the world has a bottom-level ontology" and "all phenomena can be reduced to this ontology," physics can present a clear appearance different from instrumentalism.

43. Due to limited space, this article does not introduce in detail the physical phenomena with complex mathematical representations, such as gravitational time dilation and the Higgs mechanism, under this 3D ontological realist reductionist perspective. Those interested may write to the author for discussion or visit the author's ORCID webpage.